

# I-Hsun Chen

## Curriculum Vitae

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### Research Interests

Random graphs, large deviations, interacting particle systems, spin glasses.

### Education

- 2021–present **Brown University**, Providence, RI  
Ph.D. Candidate, Applied Mathematics.  
Expected Graduation Date: May 2027.
- 2024 **Brown University**, Providence, RI  
ScM in Applied Mathematics.
- 2016–2020 **National Taiwan University**, Taipei, Taiwan  
B.S. in Mathematics.

### Research Experience

- 2021–present **Ph.D. Student**, Brown University, Providence, RI  
Advisor: Kavita Ramanan.
- Summer 2021 **Research Assistant**, National Center for Theoretical Sciences, Taipei, Taiwan
- 2020–2021 **Research Assistant**, Academia Sinica, Taipei, Taiwan  
Advisors: Ju-Yi Yen, Chii-Ruey Hwang.  
Excursion Approach to Central Limit Theorem for Some Null Recurrent Diffusion Processes.
- 2019–2020 **Undergraduate Research**, National Taiwan University, Taipei, Taiwan  
Advisor: Chii-Ruey Hwang.  
Accelerating Markov Chain Monte Carlo method through stochastic differential equations driven by Lévy processes.

### Publications

1. **I-Hsun Chen**, Ivan Lee, Kavita Ramanan, Sarath Yasodharan. “Gibbs Conditioning, Atypical Consensus and Splitting Gibbs Measures on Random Regular Graphs.” Accepted to *Mathematical Physics, Analysis and Geometry*. (special issue in honor of Claudio Landim's 60th birthday.) arXiv:2603.01322
2. **I-Hsun Chen**, Kavita Ramanan, Sarath Yasodharan. “Large Deviation Principles for Neighborhood Empirical Measures of Contextual Stochastic Block Models.” Preprint, 2026.
3. **I-Hsun Chen**, Kavita Ramanan. “Large Deviation Principles for Component Empirical Measures of Colored Configuration Models” In preparation.
4. **I-Hsun Chen**, Kavita Ramanan. “Large Deviation Principles for Erdős-Rényi and Unbounded Degree Configuration Models with i.i.d. Discrete Marks” In preparation.

### Conferences and Workshops

- Mar 2026 **Seminar on Stochastic Processes**  
Lightening talk and Poster: Gibbs conditioning and atypical consensus on random regular graphs
- Dec 2024 **BIRS-CMI Research School: Mathematical Foundations of Network Models and Their Applications**  
Student lead of the project: Bethe Prediction for Statistical Physics Models with Non-pairwise Interactions
- Aug 2024 **Princeton Machine Learning Theory Summer School**

May 2023 **Graphical Models, Statistical Inference, and Algorithms**  
Jun 2022 **PIMS-CRM Summer School in Probability**

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## Invited Talks

July 2025 **Large Deviations of the Neighborhood Empirical Measure of Heterogeneous Marked Sparse Random Graphs**, *Mathematical Congress of the Americas 2025*, Miami, FL  
June 2025 **A Sanov-type Theorem for Marked Sparse Random Graphs**, *INFORMS APS 2025*, Atlanta, GA  
December 2024 **Large Deviations of the Empirical Measure of Heterogeneous Sparse Graphs with Polish Spaced Valued Marks**, *BIRS-CMI workshop "Mathematical foundations of network models and their applications"*, Chennai, India  
2023 **A Sanov-type Theorem for Unimodular Marked Random Graphs and Its Applications to Spin Models**, *NTU Math Colloquium*, Taipei, Taiwan  
2021 **Brownian Additive Functionals Averaged by Local Time**, *Ritsumeikan University Mathematical Finance Seminar*, Kyoto, Japan. (Virtual)

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## Awards

2024-2026 **Government Scholarship to Study Abroad**, *Ministry of Education*, Taiwan (R.O.C.)  
2019 **Scholarship for Excellence**, *National Taiwan University*  
2017-2019 **Sun-Yung A. Chang & Paul C. Yang Scholarship**, *National Taiwan University*  
2019 **The 10th S.-T. Yau College Student Mathematics Contests**, *Honorable Mention in Probability and Statistics*  
2015 **Asian Pacific Mathematical Olympiad**, *Honorable Mention*

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## Advising and Mentoring

2025-2026 **Ritvik Teegaravapu**, *California Institute of Technology*, WAVE Fellows Program  
Gibbs Conditioning Principles for Dynamical Models on Sparse Random Graphs  
Co-advisor, with Kavita Ramanan.  
2023-2025 **Ivan Lee**, *Brown University*, Undergraduate Research and Honor Thesis  
Gibbs Conditioning Principles for Marked Erdős-Rényi and Random Regular Graphs  
Co-advisor, with Kavita Ramanan and Sarath Yasodharan.  
2023-2024 **Yizhong Hu**, *Brown University*, Undergraduate Research and Honor Thesis  
Annealed Pressure of Nearest-Neighbor Non-pairwise Interaction Spin-Glass Models on Random Regular Graphs (J.L. Stein award for undergraduate excellence in applied mathematics)  
Co-advisor, with Kavita Ramanan and Sarath Yasodharan.

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## Service and Teaching

2022 **APMA Directed Reading Program**, *Brown University*  
Mentor for "Mean-Field Models in Spin Glasses"  
2020 **Summer Student Seminar**, *National Taiwan University*  
Co-organizer and Lecturer of the Analysis Section  
**Teaching Assistant**, *Brown University*  
Operation Research: Probabilistic Method (Spring 2023), Statistical Inference (Fall 2022)  
**Teaching Assistant**, *National Taiwan University*  
Stochastic Calculus  
**Teaching Assistant**, *National Taiwan University*  
Calculus(II) (Spring 2020), Calculus(I) (Fall 2019), General Calculus (Spring 2019, Fall 2018)  
**Member**, *Institute of Mathematical Statistics (IMS)*, *American Mathematical Society (AMS)*  
**Reviewer**, *Annals of Applied Probability*, *Stochastic Processes and Their Applications*